

Chapter 3: Dynamics

Key Concepts

Force — An agent that changes or tends to change the state of rest or uniform motion of a body. It is a vector quantity. SI unit: newton (N).

Inertia — The natural tendency of a body to resist any change in its state of rest or uniform motion. Greater mass means greater inertia.

Newton's First Law (Law of Inertia) — A body at rest stays at rest, and a body in uniform motion stays in uniform motion, unless acted upon by an unbalanced external force.

Mass — The measure of inertia of a body. Larger mass is harder to start moving or stop. SI unit: kilogram (kg).

Newton's Second Law — When an unbalanced force acts on a body, it produces acceleration in the direction of the force, directly proportional to the force and inversely proportional to the mass. Mathematically: $F = ma$.

Newton's Third Law — For every action, there is an equal and opposite reaction. Action and reaction act on different bodies, so they do not cancel each other.

Momentum — The quantity of motion possessed by a moving body. It is the product of mass and velocity. Vector quantity. Formula: $p = mv$. SI unit: $\text{kg}\cdot\text{m/s}$ or $\text{N}\cdot\text{s}$.

Impulse — The product of force and the time interval for which it acts. It equals the change in momentum. Formula: $\text{Impulse} = F \times t = \Delta p$. SI unit: $\text{N}\cdot\text{s}$ or $\text{kg}\cdot\text{m/s}$.

Law of Conservation of Momentum — In the absence of an external unbalanced force, the total momentum of an isolated system remains constant. For two bodies: $m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$.

Friction — The force that opposes the relative motion between two surfaces in contact. It acts parallel to the surfaces and opposite to the direction of motion.

Types of Friction — Friction between surfaces at rest relative to each other. Maximum just before motion starts.

- **Static Friction** — Friction when one surface slides over another. It is less than static friction.
- **Sliding Friction** — Friction when one surface rolls over another. It is the smallest of the three.

Advantages of Friction — Friction is possible due to friction between shoes and ground.

- Vehicles can move on roads due to friction between tires and road.
- Writing on paper is possible due to friction.
- Nails and screws stay in place due to friction.

Disadvantages of Friction — Friction causes wear and tear in machine parts.

- Produces unwanted heat — engines need cooling.
- Wastes energy — moving parts need extra force to overcome friction.
- Slows down moving objects.

Methods of Reducing Friction.

- Using ball bearings or roller bearings.
- Streamlining shapes to reduce air resistance.
- Polishing surfaces.

Uniform Motion — When a body covers equal distances in equal intervals of time. Net force = 0.

Non-Uniform Motion — When a body covers unequal distances in equal intervals of time. Net force $\neq 0$.

MCQ's

1. Newton's first law of motion is also called:

A. Law of momentum B. Law of inertia C. Law of acceleration D. Law of reaction

Answer: B

2. The SI unit of force is:

A. Joule B. Newton C. Pascal D. Watt

Answer: B

Answer: C

A. Rolling friction B. Sliding friction C. Static friction D. Fluid friction

Answer: C

A. m/v B. mv C. ma D. m/a

Answer: C

A. Newton's first law B. Newton's second law C. Newton's third law D. Law of inertia

Answer: C

A. Mass only B. Velocity only C. Both mass and velocity D. Force only

Answer: C

A. $Ft = p$ B. $Ft = 9G$ C. $F = ma$ D. $p = mv$

Answer: B

A. High friction B. Low friction C. No friction D. Static friction

Answer: B

A. Increase friction B. Reduce friction C. Increase weight D. Reduce weight

Answer: B

A. Shape B. Volume C. Mass D. Density

Answer: C

A. Walking B. Braking C. Wear of machine parts D. Writing

Answer: C

A. External force acts B. No external force acts C. Friction is present D. Motion is circular

Answer: B

A. Maximum B. Infinity C. Zero D. $1 \text{ kg}\cdot\text{m/s}$

Answer: C

A. Static friction B. Sliding friction C. Rolling friction D. Fluid friction

Answer: C

[illegible]

Short Questions

Q1. State Newton's first law of motion.

Ans: Newton's first law states that a body at rest will remain at rest, and a body in uniform motion will continue its uniform motion in a straight line, unless an unbalanced external force acts on it. This property is called inertia.

Q2. What is the relation between force, mass, and acceleration?

Ans: According to Newton's second law, the acceleration produced in a body is directly proportional to the applied unbalanced force and inversely proportional to the mass of the body. Mathematically: $F = ma$, where F is force in newtons, m is mass in kg, and a is acceleration in m/s^2 .

Q3. Explain Newton's third law with an example.

Ans: Newton's third law states that for every action, there is an equal and opposite reaction. Example: When a person jumps from a boat onto the shore, the person pushes the boat backward (action), and the boat pushes the person forward (reaction), enabling the jump.

Q4. Define momentum and write its formula.

Ans: Momentum is the quantity of motion possessed by a moving body. It is the product of mass and velocity. Formula: $p = mv$ (where p = momentum in $\text{kg}\cdot\text{m/s}$, m = mass in kg , v = velocity in m/s). It is a vector quantity whose direction is same as velocity.

Q5. Define impulse. How is it related to momentum?

Ans: Impulse is the product of force and the time interval during which the force acts. Mathematically: Impulse = $F \times t$. According to the impulse-momentum equation, impulse equals the change in momentum: $F \times t = m v_f - m v_i$.

Q6. State the law of conservation of momentum.

Ans: The law of conservation of momentum states that in the absence of an external unbalanced force, the total momentum of an isolated system remains constant. For two colliding bodies: $m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2$, where u and v are initial and final velocities respectively.

Q7. Differentiate between static friction and sliding friction.

Ans: Static friction acts between surfaces at rest relative to each other. It is the maximum frictional force and must be overcome to start motion. Sliding friction acts when one surface slides over another. Sliding friction is less than static friction — it takes less force to keep an object moving than to start it moving.

Q8. List two methods to reduce friction.

Ans:

1. Using lubricants (oil, grease) between surfaces — lubricant fills microscopic irregularities and reduces contact.
2. Using ball bearings or roller bearings — converts sliding friction into rolling friction, which is much smaller.

Q9. Why does a person fall forward when a moving bus stops suddenly?

Ans: When the bus suddenly stops, the lower part of the person (in contact with the bus) stops with the bus. However, the upper part of the body continues to move forward due to inertia of motion. This causes the person to fall forward.

Q10. Of a truck and a car moving at the same speed, which has more momentum and why?

Ans: The truck has more momentum. Since momentum $p = mv$, and both vehicles have the same velocity v , the truck's larger mass gives it greater momentum. This is why it is harder to stop a truck than a car moving at the same speed.

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Long Questions

Q1. State and explain Newton's laws of motion with examples.

Ans:

First Law (Law of Inertia): A body at rest stays at rest, and a body in uniform motion stays in uniform motion, unless acted upon by an unbalanced force.

- Example: A book on a table stays at rest until someone pushes it.
- Example: A passenger lurches forward when a bus suddenly stops — body continues moving due to inertia.

Second Law: When an unbalanced force acts on a body, it produces acceleration proportional to the force and inversely proportional to the mass. $F = ma$ ($1 \text{ N} = 1 \text{ kg}\cdot\text{m/s}^2$).

- Example: Pushing an empty shopping cart is easy (small m , large a). Pushing the same cart full of groceries is harder (large m , small a for same force).

Third Law: For every action, there is an equal and opposite reaction. Action and reaction act on different bodies.

- Example: A rocket pushes exhaust gases downward (action); the gases push the rocket upward (reaction).
- Example: While walking, we push the ground backward; the ground pushes us forward.

Q2. Explain the concept of friction. Discuss its types, advantages, and disadvantages.

Ans:

Friction is a force that opposes relative motion between two surfaces in contact. It always acts parallel to surfaces and opposite to the direction of motion.

Types:

Static Friction — Maximum friction; must be overcome to start motion.

Sliding Friction — Acts during sliding; smaller than static friction.

Rolling Friction — Acts during rolling motion; smallest of all three.

Advantages:

- Enables walking and running without slipping.
- Allows vehicles to move and brake on roads.
- Makes writing and drawing possible.
- Keeps knots tied and nails fixed in walls.

Disadvantages:

- Causes wear and tear of machine parts.
- Produces unwanted heat in machinery.
- Wastes energy — extra fuel needed to overcome friction.
- Slows down moving objects.

Q3. Derive the relationship between force and momentum. Show that $F = ma$ is a special case of the impulse-momentum equation.

Ans:

Momentum is defined as $p = mv$.

Rate of change of momentum:

$$\frac{\Delta p}{\Delta t} = \frac{mv_f - mv_i}{t}$$

$$\frac{\Delta p}{\Delta t} = \frac{m(v_f - v_i)}{t}$$

$$\frac{\Delta p}{\Delta t} = m \times \frac{v_f - v_i}{t}$$

$$\frac{\Delta p}{\Delta t} = m \times a$$

Newton's second law states that force equals rate of change of momentum:

$$F = \frac{\Delta p}{\Delta t} = ma$$

The impulse-momentum equation is $F \times t = \Delta p$. When we divide both sides by t , we get $F = \frac{\Delta p}{t}$. Thus, $F = ma$ is a special case where we consider the instantaneous rate of change of momentum.

Q4. State and prove the law of conservation of momentum. Give an example.

Ans:

Statement: In the absence of an external unbalanced force, the total momentum of an isolated system remains constant. Momentum may be transferred between bodies, but it is never created or destroyed.

Proof (for two colliding bodies):

Consider two bodies A and B with masses m_1 and m_2 . Let their initial velocities be u_1 and u_2 . They collide, and after collision their velocities become v_1 and v_2 . Let the collision last for time t .

Force on A due to B:

$$F_{AB} = \frac{m_1(v_1 - u_1)}{t}$$

Force on B due to A:

$$s \bullet = v \bullet t + \frac{1}{2} a t^2 = 0 + \frac{1}{2}(4)(16) = 32 \text{ m}$$

Phase 2 (no force): $v = 16 \text{ m/s}$ (uniform), $t_2 = 10 - 4 = 6 \text{ s}$

$$s_2 = v \times t_2 = 16 \times 6 = 96 \text{ m}$$

$$\text{Total distance: } s = s_1 + s_2 = 32 + 96 = 128 \text{ m}$$